NIMAR Sentiment Analysis API Documentation

Overview

The NIMAR Sentiment Analysis API provides sentiment classification capabilities using the gpt-oss-20b model. This API allows you to analyze text and determine its emotional tone or sentiment.

Base URL: `https://sentiment-api-dgx.nimar.gov.pk/`

Version: v1

Authentication

All sentiment analysis requests require authentication using an API key passed in the request header.

Header:

```
x-api-key: nimar321
```

Endpoints

1. Root Endpoint

GET `/`

Simple root endpoint to verify API accessibility.

Response: `200 OK`

---

2. Health Check

GET `/health`

Verifies the health status of the API by checking the GPT-OSS model availability.

Response: `200 OK`

Example Request:

```bash
```

```
curl -X GET "https://sentiment-api-dgx.nimar.gov.pk/health"
```

---

3. Analyze Sentiment

POST `/v1/sentiment`

Analyzes the sentiment of the provided text input.

Headers:

- `x-api-key` (required): `nimar321`
- `Content-Type`: `application/json`

Request Body:

Field	Type	Required	Description
text	string	Yes	The input text for sentiment analysis

Example Request:

```bash
curl -X POST "https://sentiment-api-dgx.nimar.gov.pk/v1/sentiment" \
  -H "x-api-key: nimar321" \
  -H "Content-Type: application/json" \
  -d '{
    "text": "This is a great product! I am very satisfied."
  }'
```

Example Request Body:

```json
{
  "text": "This is a great product! I'm very satisfied with my purchase."
}
```

Response: `200 OK`

Returns sentiment classification results for the provided text.

Error Responses:

- `422 Unprocessable Entity`: Validation error (e.g., missing required fields)

Example Error Response:

```json
{
  "detail": [
    {
      "loc": ["body", "text"],
      "msg": "field required",
      "type": "value_error.missing"
    }
  ]
}
```

Usage Examples

Python

```python
import requests

url = "https://sentiment-api-dgx.nimar.gov.pk/v1/sentiment"
headers = {
    "x-api-key": "nimar321",
    "Content-Type": "application/json"
}
data = {
    "text": "This product exceeded my expectations!"
}

response = requests.post(url, json=data, headers=headers)
print(response.json())
```

JavaScript (Node.js)

```javascript
```

```
const axios = require('axios');

const url = 'https://sentiment-api-dgx.nimar.gov.pk/v1/sentiment';
const headers = {
  'x-api-key': 'nimar321',
  'Content-Type': 'application/json'
};
const data = {
  text: 'This product exceeded my expectations!'
};

axios.post(url, data, { headers })
  .then(response => console.log(response.data))
  .catch(error => console.error(error));
```

JavaScript (Fetch API)

```javascript
fetch('https://sentiment-api-dgx.nimar.gov.pk/v1/sentiment', {
  method: 'POST',
  headers: {
    'x-api-key': 'nimar321',
    'Content-Type': 'application/json'
  },
  body: JSON.stringify({
    text: 'This product exceeded my expectations!'
  })
})
  .then(response => response.json())
  .then(data => console.log(data))
  .catch(error => console.error('Error:', error));
```

cURL

```bash
curl -X POST "https://sentiment-api-dgx.nimar.gov.pk/v1/sentiment" \
  -H "x-api-key: nimar321" \
  -H "Content-Type: application/json" \
  -d '{"text": "This product exceeded my expectations!"}'
```

PHP

```php
<?php
$url = 'https://sentiment-api-dgx.nimar.gov.pk/v1/sentiment';
$data = array('text' => 'This product exceeded my expectations!');

$options = array(
    'http' => array(
        'header'  => "Content-type: application/json\r\n" .
                     "x-api-key: nimar321\r\n",
        'method'  => 'POST',
        'content' => json_encode($data)
    )
);

$context  = stream_context_create($options);
$result = file_get_contents($url, false, $context);
echo $result;
?>
```

Error Codes

Status Code	Description
200	Successful response
422	Validation error - check request body format
401	Unauthorized - invalid or missing API key
500	Internal server error

Best Practices

1. **Text Length**: While not specified in the API, keep text inputs reasonable in length for optimal performance
2. **Rate Limiting**: Be mindful of potential rate limits (contact NIMAR for specific limits)
3. **Error Handling**: Always implement proper error handling in your application
4. **API Key Security**: Keep your API key secure and do not expose it in client-side code or public repositories

---

Support

For API access, technical support, or questions, please contact NIMAR.

Organization: NIMAR (National Institute of Maritime Affairs) **Website:** nimar.gov.pk